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Paleomagnetic evidence of the rotation of central Japan and the paleoposition of Japan

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Abstract

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Paleomagnetic studies of the Daishoji, the Echigo-Yuzawa and the Kitaimogawa areas revealed complex intra-block tectonic rotations in the Central Honshu block which is sandwiched in between the Southwest and Northeast Honshu blocks.

Paleogeographic maps of 16, 15 and 14 Ma are reconstructed by using paleomagnetic declinations obtained so far in Japan.

Introduction

The tectonic bending of Japan was first proposed by Kawai et al. (1961, 1962) on the basis of paleomagnetic measurements of rocks collected from all over the Japanese main island of Honshu. Paleomagnetic declinations of the pre-Neogene igneous and sedimentary rocks in the southwestern Honshu deflect greatly to the east while those in the northwestern part of the island deviate to the west. This fact shows a very clear contrast with the evidence that the younger Neogene volcanic rocks both in southwestern and northeastern Honshu have the same north-seeking declinations. Kawai et al. (1961) concluded, consequently, that the difference in declination of pre-Neogene age was caused by the counterclockwise rotation of northeastern Honshu relative to the southwestern part. They assumed that the rotation occurred at some time between the Cretaceous and the Neogene. Kawai et al. (1969, 1971) verified the temporal sequence of the rotation of northeastern Honshu by the changes in paleomagnetic declination of granitic bodies with known ages.

Otofuji and Matsuda (1983, 1984) revealed that the easterly-deviated declination can be traced up to the Early Miocene by their paleomagnetic study of acidic volcanic rocks combined with fission-track dating in southwestern Honshu. Torii et al. (1985) and Otofuji et al. (1985a) suggested that the rotation of the southwestern Honshu Island was caused by the opening of the Japan Sea and that it was completed within a very short time which started at the latest Early Miocene and was finished by the beginning of Middle Miocene. Southwestern Honshu rotated clockwise about 58° (Otofuji and Matsuda, 1983, 1984) while the northeastern part of the island underwent a counterclockwise rotation of about 50° (Otofuji et al., 1985b,c).

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Hirooka et al. (1986) clarified, however, that the tectonic motion of the central part of the island cannot be described by a simple bending of the island. Especially, since the region along the Itoigawa-Sizuoka Tectonic Line (ISTL), which is the Neogene boundary of the geologic provinces of Northeast and Southwest Japan, is not considered to be a single block which belongs to southwestern or northeastern Honshu and which underwent a clockwise or a counterclockwise rotation as a whole, but rather to be divided into several sub-blocks, each of which rotated differently. Itoh (1986) studied the Yatsuo Formation of the uppermost Lower Miocene sedimentary sequences which is located about 50 km west of the ISTL. He could not find such an extreme easterly declination as was reported from Southwestern Honshu. The mean declination of the Yatsuo Formation is about 20°E. The Yatsuo Formation does not belong to the Southwest Honshu block. Nakajima and Hirooka (1986) confirmed that the volcanic rocks and tuff layers of late Early Miocene age in the Fukui Prefecture show very deflected declinations of about 40°E. The most eastward extension of the Southwestern Honshu block, therefore, can be traced about 150 km west of the ISTL. The eastern boundary of the Southwest Honshu block must be in the east of the Fukui Prefecture and in the west of the Yatsuo region.

The Tanakura Tectonic Line (TTL) is generally considered to be the pre-Neogene geotectonic boundary between Northeast and Southwest Japan. Since the paleomagnetic studies of the regions which are located east on the TTL indicate that the regions rotated counterclockwise by about 50° (Otofuji et al., 1985b; Hamano and Tosha, 1985), the TTL very probably is the western boundary of the Northeast Honshu block. The extent of the central part of the Honshu Island discussed here, therefore, should be defined by the region in between the TTL and the Southwestern Honshu block.

Paleomagnetic studies have been carried out in the areas of Kita-Imogawa, Echigo-Yuzawa and Daishoji to determine the intrablock tectonic motion within the Central Honshu block as well as to reconstruct the paleogeography of Japan in Miocene time.

Paleomagnetic sampling and geological settings of sampling sites

Paleomagnetic sampling was carried out at the above-mentioned three areas in the central Honshu, i.e. the Daishoji area, located west of the ISTL, and the Kita-Imogawa and the Echigo-Yuzawa areas located east of the tectonic line as indicated in Fig.1. About ten hand samples were collected from each site.

The Daishoji area

According to the geology of this area studied by Kaseno et al. (1955), Bito et al. (1980), Ogasawara et al. (1981) and Tsuchi et al. (1981), the Early Miocene Series is divided into four formations which are the Kayano, the Yamanaka, the Kawaminami and the Hosotsubo

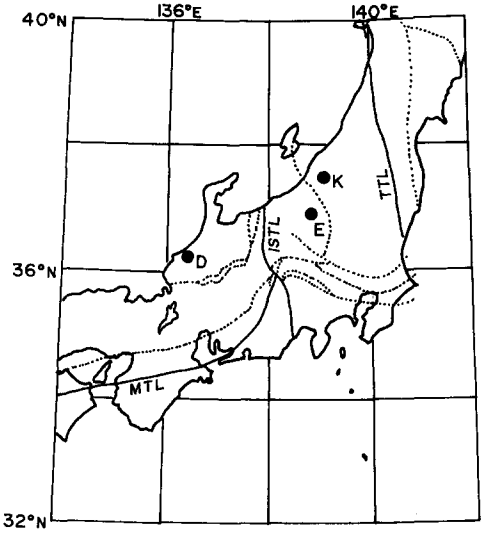


Fig.1. Map showing the localities of paleomagnetic sampling sites together with the major geotectonic lines and the boundaries of geologic provinces (dotted lines). *D*=Daishoji area. *E*=Echigo-Yuzawa area. *K*=Kita-Imogawa area. *MTL*=Median Tectonic Line. *ISTL*=Itoigawa-Shizuoka Tectonic Line. *TTL*=Tanakura Tectonic Line.

Formations in ascending order. The Kawaminami Formation consists of four members: the Uneichinono Dacite, the Bessho Rhyolite, the Sosori Sandstone and the Sou Tuff. The Hosotsubo Formation is considered to belong to the N 8 and N 9 zones of Blow's zonation (Blow, 1969) from planktonic foraminiferal study (Bito et al., 1980).

Paleomagnetic sampling was carried out at three horizons in the Yamanaka Formation (Sites INS-32, 31 and 30), three horizons in the Uneichinono Dacite Member (Sites INS-0, 34 and 35), one horizon in the Sosori Sandstone Member (Site INS-2) and one horizon in the Hosotsubo Formation (Site INS-3).

The Kita-Imogawa area

An Early to Middle Miocene sedimentary formation called the Nanatani Formation occurs in this area (Chihara et al., 1977). Paleomagnetic sampling was done at 11 sites along the stratigraphic sequence from the upper to the lower horizons (Sites NMS-1 to 11) in this area. Planktonic foraminiferal studies indicate that this formation correlates with the N 8 and N 9 zones (Matsuoka, 1979; Tsuchi et al., 1979). Paleomagnetic sampling was done at the upper horizons of this sequence, so that the samples were of a N 9 age.

The stratigraphic horizons of the paleomagnetic sampling sites of the Daishoji and the Kitaimogawa areas are shown in Fig.2.

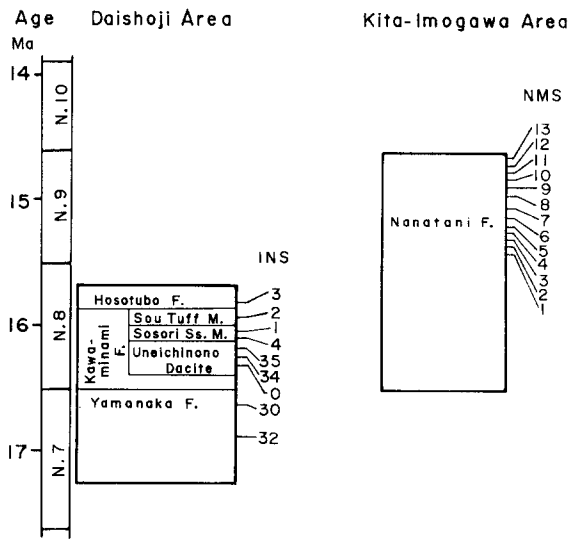


Fig.2. Stratigraphic horizons of the paleomagnetic sampling sites in the Daishoji and Kita-Imogawa areas.

The Echigo-Yuzawa area

According to the geological map of the Echigo-Yuzawa district published by the Geological Survey of Japan (Chihara et al., 1981), Paleogene granitic rocks and Neogene acidic and basic dikes which intruded into the above-mentioned granites, the Neogene sediments and younger granitic intrusives of 5.9 Ma (Kawano and Ueda, 1966) occur in this area. Paleomagnetic sampling was done from Paleogene granitic rocks at three sites (EY-4, 6 and 10), from basic dikes at ten sites (EY-1, 3, 5, 7, 12, 13, 14, 15, 16 and 17), from acidic dikes at four sites (EY-2, 8, 9 and 11), from the Early Miocene sedimentary sequence called the Matsukawa Formation at one site (EY-19) and from the younger granitic intrusive body at one site (EY-0).

Figure 3 shows the stratigraphic horizons of sampling sites of this area.

Paleomagnetic measurements

Sample preparation and paleomagnetic measurements were carried out in the following procedure.

First, two or three specimens were cored from each hand sample. For each site, we selected one specimen from every hand sample and put them together into a specimen set. One set consists of about ten specimens so that we could provide two or three sets for each site. After measurement of the natural remanent magnetization (NRM) of all the specimens, three or four pilot specimens from a set were selected for each site to be submitted to stepwise alternating field (a.f.) demagnetization. The demagnetization was carried out at steps of 5 mT interval. The maximum field of demagnetization was up to 50 mT at the lowest and occasionally up to 80 mT. An appropriate

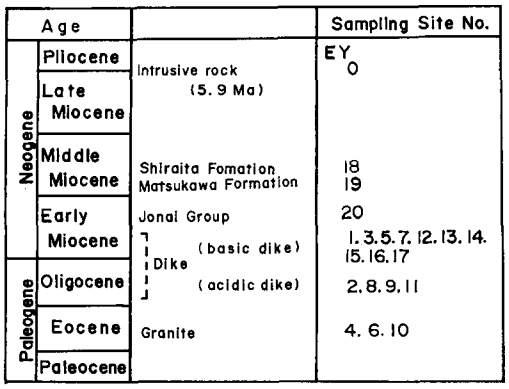


Fig.3. Stratigraphic horizons of the paleomagnetic sampling sites in the Echigo-Yuzawa area.

demagnetizing step was defined for each site when the scatter of magnetic directions of the pilot specimens became minimal. Then, all remaining specimens of the set were demagnetized at the appropriate step and neighboring two steps. We defined the field of the step where the Fisher's precision parameter (Fisher, 1953) of the set attained maximum value for the optimum demagnetizing field (ODF), and we took the mean declination and inclination of the step as paleomagnetic data of the site.

For each site, two or more pilot specimens were also selected from the second set of the same site for submitting to the thermal demagnetization. Ten to fourteen temperature steps, for example, 100, 200, 300, 350, 400, 450, 500, 540, 570 and 600°C were applied to confirm the thermal stability of remanence of specimens. According to the variation pattern of the magnetic direction and intensity, the temperature intervals for the steps were changed for each site to apply the most suitable demagnetizing interval.

In the case that the thermally treated remanent directions of pilot specimens of a site changes differently from those of a.f. demagnetization, all the remaining specimens of the second set were thermally demagnetized at three or more steps which were assumed to be around the most appropriate step. The mean declination and inclination of a step at which Fisher's precision parameter K showed the maximum value were cited as the paleomagnetic data of the optimum demagnetizing temperature (ODT) for the site.

The results of the paleomagnetic measurements are listed in Table I. In the table, D and I are in situ declination and inclination, and Dc and Ic are tilt corrected ones. It clearly shows from the table that the tilt corrected mean declinations of the Daishoji and the Echigo-Yuzawa areas deviate eastward, while those of the Kita-Imogawa area westward.

In calculating the area mean directions, the value of Site EY-15 which greatly deviated from other sites in the area was excluded because its direction of remanence is not considered to be the record of the ordinary geomagnetic field.

Discussion and conclusion

Southwest Japan underwent a very rapid clockwise rotation at around 15 Ma as clarified by Otofuji et al. (1985b). In contrast, Northwest Honshu seems to have rotated counterclockwise according to paleomagnetic data obtained so far (Otofuji et al., 1985b: Hamano and Tosha, 1985). The central part of the Honshu such as the Chubu and the Kanto districts is, therefore, supposed to have suffered from a very complex geotectonism because that part is sandwiched between northeast and southwest Honshu, each undergoing a tectonic rotation contrary to the other. The eastward declination of the Daishoji area indicates that this area belongs to the southwest Honshu block which rotated as a single block. On the contrary, the present paleomagnetic results of the declination for the Echigo-Yuzawa area deviates greatly eastward, while that of the Kita-Imogawa area deviates westward, in spite of the fact that both areas are located east of the ISTL. This fact indicates that a very complex intra-block tectonism occurred in the region.

Moreover, the extremely deviating declination of 93.7°E of the Chichibu Tertiary System in the Kanto Mountains (Hyodo and Niitsuma, 1986) which is located east of the ISTL in the central part of the Honshu indicates that this

TABLE I

Paleomagnetic results of the Daishoji, the Kita-Imogawa and the Echigo-Yuzawa areas

Site		*N*	*D*	*I*	α95	*K*	*D*c	*I*c	Lat.	Long.	ODF/ODT
Daishoji area											
INS	32	11	213.2	−67.2	2.5	340	217.7	−51.4	36°14.6′	138°18.5′	10 mT
INS	3	13	208.3	−67.1	3.7	125	225.1	−45.5	36°14.7′	138°18.4′	10 mT
INS		8	242.4	−69.4	8.1	48	231.9	−58.5	36°15.2′	136°13.4′	12.5 mT
INS	34	9	224.3	−64.4	4.3	143	218.1	−50.9	36°15.3′	136°18.3′	12.5 mT
INS	35	11	203.9	−61.9	3.9	137	218.1	−37.3	36°15.4′	136°18.2′	12.5 mT
INS	3	10	245.2	−49.9	11.7	18	216.4	−51.9	36°16.6′	136°18.6′	12.5 mT
Mean		6			6.7	101	220.9	−49.4			
Kita-Imogawa area											
NMS	3	8	11.2	58.9	15.2	14	−27.3	49.6	39°31.8′	139°08.2′	5 mT
NMS	4	9	6.7	62.8	10.0	27	−20.2	47.0	39°31.9′	139°08.1′	5 mT
NMS	7	7	−12.6	51.4	16.8	14	−25.8	38.7	39°31.9′	139°08.0′	100 C
NMS	10	4	10.2	54.7	21.3	20	−23.0	52.3	39°32.0′	139°07.9′	100 C
Mean		4			7.1	169	−24.1	47.0			
Echigo-Yuzawa area											
EY	0	10	1.3	58.1	7.9	39	1.3	58.1	36°51.9′	138°48.9′	5 mT
EY	19	8	49.2	75.5	15.2	14	−14.1	51.8	36°53.5′	138°48.3′	400 C
EY	7	9	233.1	−21.0	11.5	21	233.1	−21.0	36°53.2′	138°48.7′	570 C
EY	13	9	225.1	−12.5	3.0	300	225.1	−12.5	36°52.8′	138°49.3′	20 mT
EY	14	10	210.9	−48.0	2.4	416	210.9	−48.0	36°52.8′	138°49.3′	35 mT
EY	15	9	228.2	10.8	9.3	32	228.2	10.8	36°52.8′	138°49.3′	500 C
EY	16	9	224.1	−56.6	11.6	21	224.1	−56.6	36°53.3′	138°48.8′	450 C
EY	17	7	213.2	−64.6	13.4	21	213.2	−64.6	36°53.2′	138°48.8′	540 C
EY	8	10	224.7	−27.0	6.3	60	224.7	−27.0	36°53.2′	138°48.7′	15 mT
EY	10	10	225.8	−45.6	9.3	28	225.8	−45.6	36°53.2′	138°48.7′	10 mT
Mean		7			15.5	16	223.7	−39.5			

N: number of sites, *D*: in situ declination, *I*: in situ inclination, α95: Fisher's angle of confidence, *K*: Fisher's precision parameter, *D* c: tilt corrected declination, *I* c: tilt corrected inclination, Lat.: latitude of sampling site (North), Long.: longitude of sampling site (East), ODF/ODT: optimum demagnetizing field or optimum demagnetizing temperature.

region was deformed and rotated extensively. Hyodo and Niitsuma (1986) suggested that this tectonic deformation was originated by both the opening of the Japan Sea and the collision of the Izu Peninsula. According to Hyodo and Niitsuma, the system rotated twice. The first rotation took place at about 15 Ma, the time when the Japan Sea opened. The Chichibu Tertiary System rotated 47° as part of the Southwestern Honshu. Then, in the late Miocene, the second rotation occurred when the Izu Peninsula collided with Honshu. The system rotated clockwise again by 45°. If this is the case, the Kanto District can be expected to have undergone severe deformation due to the collision. Hirooka et al. (1985) suggested that the collision occurred at around 3–5 Ma from the paleolatitude change of the Izu Peninsula.

Tertiary paleomagnetic directions hitherto obtained in Japan are listed in Table II. Declinations and inclinations in the table are the site mean directions for the areas. The means are calculated after inverting the reversely magnetized directions into the normal ones. Figure 4a–c shows paleomagnetic directions of the areas at times older than 16 Ma, around 15 Ma and younger than 14 Ma, respectively. It is evident from Fig.4a that the declinations of Southwestern Honshu deviate strongly eastward and those of Northeastern Honshu westward.

Paleomagnetic reconstruction of the paleo-

TABLE II

Paleomagnetic data of the Japanese Tertiary

Area	*N*	*D*	*I*	α95	*K*	Age	Reference
Older than 16 Ma							
Shimane	30	61.9	43.6	6.6	17	Oligo.~E. Mio.	1
Tonosho	4	42.2	60.3	11.1	70	E. Mio.	2
Ichishi F.	11	45.1	48.8	11.9	36	E. Mio.	3
Morozaki	5	58.0	56.4	12.9	36	E. Mio.	4
Mizunami	5	55.1	44.1	8.9	76	E. Mio.	4
Niu Mt., Fukui	22	44.0	49.8	11.4	8.3	18.6~15.0 Ma	5
Daishoji	6	40.9	49.4	6.7	101	E. Mio.	this paper
Sidara	4	48.0	50.0	14.0	44	E. Mio.	2
Yatsuo	2	18.1	45.2	—	—	E. Mio.	6
Noto	6	−4.8	70.9	15.9	19	27.9~16.0 Ma	7
Echigo-Yuzawa	7	43.7	39.5	15.5	16	Oligo.~E. Mio.	this paper
Sado	5	−27.1	59.9	11.1	69	28.6~19.3 Ma	7
Fukushima	4	−14.3	52.8	8.2	128	23.0~21.3 Ma	7
Yuzawa	4	−43.4	64.5	24.2	15	25~22 Ma	8
Kita-Asahi	1	−27.4	50.5	—	—	21 Ma	8
Oga	2	−30.0	47.6	—	—	21.5~20 Ma	8
Ninohe	1	−66.9	39.4	—	—	32 Ma	8
Chichibu	11	93.7	52.0	8.3	31	E.~M. Mio.	9
15±1 Ma							
Tsuyama	4	23.5	37.6	11.4	66	M. Mio.	2
Uchiura	3	−3.3	47.5	24.2	27	15.2±2.3 Ma	5
Morozaki	7	28.4	53.3	13.6	21	M. Mio.	4
Mizunami	2	25.8	56.6	—	—	M. Mio.	4
Sidara (dike)	9	18.7	68.0	9.4	31	15.1±0.5 Ma	11
Niu Mt., Fukui	3	−7.7	42.6	28.2	20	15.4±2.0 Ma	5
Yatsuo	2	6.7	48.7	—	—	M. Mio.	6
Noto	2	41.4	50.2	—	—	15.2±0.4 Ma	7
Echigo-Yuzawa	1	−14.1	51.8	—	—	M. Mio.	this paper
Kita-Imogawa	4	−24.1	47.0	7.1	169	M. Mio.	this paper
Taiheizan	2	−10.1	51.1	—	—	14.1 Ma	8
Younger than 14 Ma							
Uchiura	5	6.5	47.0	21.5	22	M.~U. Mio.	5
Niu Mt., Fukui	8	−4.3	49.1	11.8	23	M.~U. Mio.	5
Yatsuo	4	−2.5	55.6	6.6	198	M.~U. Mio.	6
Noto	1	−0.2	48.7	—	—	8.6±0.6 Ma	7
Sado	1	−24.5	51.4	—	—	13.1±2.0 Ma	7
Niigata	9	3.8	52.4	4.5	132	13.6~12 Ma	7
Echigo-Yuzawa	1	1.3	58.1	—	—	5.9 Ma	this paper
Minamiaizu-Koriyama	4	−3.8	56.7	6.7	188	13.7~10.1 Ma	7
Zao	2	−14.5	69.7	—	—	11.3 Ma	8
Shimokura (dike)	15	−18.0	62.1	5.2	54	8 Ma	10
Oga (dike)	13	−2.8	58.3	8.0	28	M. Mio.	12

N: number of sites, *D*: mean declination, *I*: mean inclination, α95: Fisher's angle of confidence, *K*: Fisher's precision parameter, 1: Otofuji and Matsuda (1984) 2: Torii (1983) 3: Hayashida and Ita (1984) 4: Hayashida (1986) 5: Nakajima and Hirooka (1986) 6: Itoh (1986) 7: Hirooka et al. (1986) 8: Otofuji et al. (1985b) 9: Hyodo and Niitsuma (1986) 10: Tsunakawa et al. (1985) 11: Tosha and Tsunakawa (1981) 12: Tosha and Hamano (1986).

geography of Japan in the Miocene can be performed by rotating blocks and sub-blocks so as to point their paleomagnetic declinations to the geographic north and to line up them parallel to each other. Reconstructed paleogeographic maps from 16 Ma, 15 Ma and 14 Ma are given in Fig.5a, 5b, and 5c respectively. The paleoposition of the Japanese Islands shown

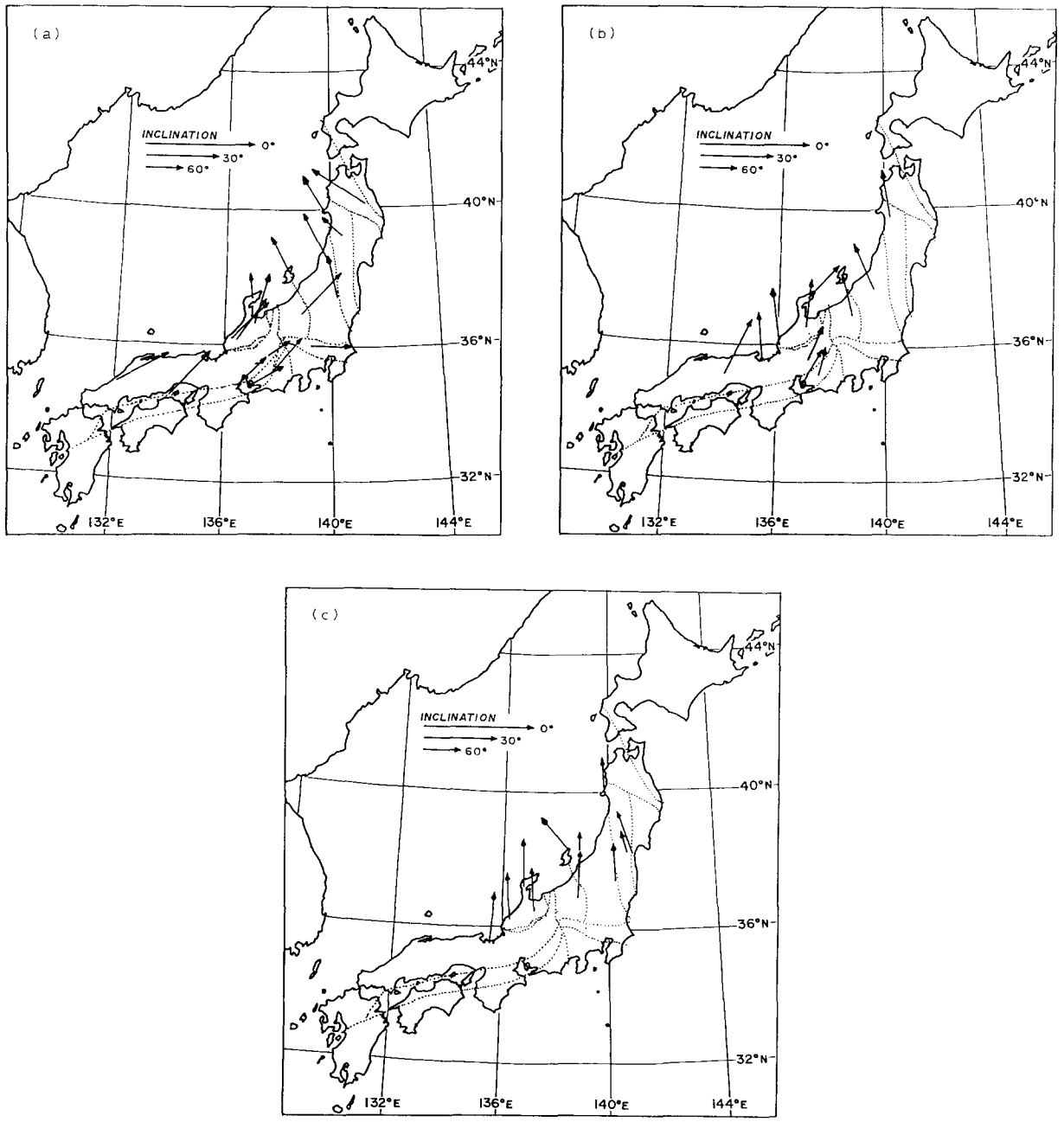


Fig.4. Paleomagnetic directions of various localities on Honshu Island. The direction of the arrows shows the paleomagnetic declination and the length of the arrows indicates the paleomagnetic inclination. a. Paleomagnetic directions for older than 16 Ma. b. Paleomagnetic directions for 15 ± 1 Ma. c. Paleomagnetic directions for younger than 14 Ma.

here is a very arbitrary one in relation to the Korean Peninsula and the Asian Continent because the paleomagnetic reconstruction is basically unable to determine the paleolongitude and the fact that the resolution to determine the paleolatitude of Japan was very low as a result of the scatter in paleomagnetic inclination values.

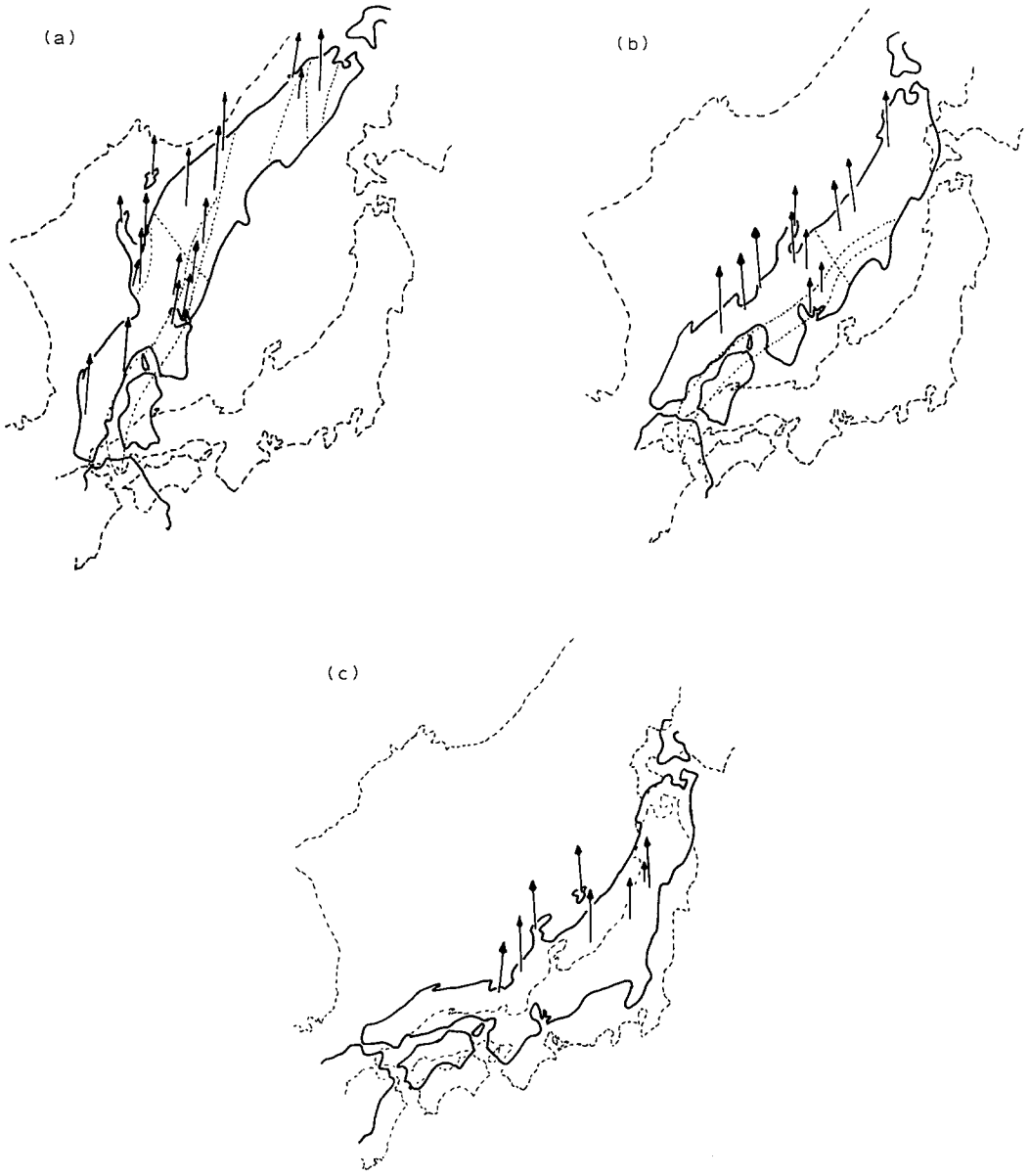


Fig.5. Paleogeographic maps of Japan at 16 Ma, 15 Ma and 14 Ma reconstructed from paleomagnetic directions. The paleopositions (paleolatitudes and paleolongitudes) relative to the Asian Continent and Japanese Islands are arbitrary.
Dotted lines drawn in the Japanese Islands indicate the major geotectonic lines and the major boundaries of geologic provinces. a. Paleogeographic map at 16 Ma. b. Paleogeographic map at 15 Ma. c. Paleogeographic map at 14 Ma.

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